

New Frontiers of UAV Civilian Applications

Applications for unmanned aerial vehicles continue to grow by leaps and bounds, especially when they involve difficult tasks or dangerous situations

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As part of the push to increase uses of civilian drones, nearly 50 companies are developing some 150 different systems, ranging from miniature models to those with wingspans comparable to airliners. Some of the manufacturers are offering do-it-yourself unmanned aerial vehicles (UAVs) like kit planes.

Present applications

Civilian uses ranging from disaster rescue missions, wildlife monitoring and firefighting are some of the many applications. The idea of drones quickly transporting life-saving parcels such as blood samples and medicines even in the areas where roads and infrastructure are poor should be especially attractive to Indian conditions.

Forest surveys conducted by a camera-equipped drone provide researchers with extensive landscape information in the form of 3D interactive maps. Ultimately, such maps will be publicly accessible via open-source computer vision software. For nuclear decommissioning or radio contamination cleanup, UAVs with radiation detection and mapping software are being used.

Configuration of different types of UAVs

The first picture on column two shows vertical take-off and landing UAV that combines the ability to hover with high-speed horizontal flight. Testing will result in fully autonomous operations. Originally designed for commercial mapping operations, the slim vehicle uses a shrouded ducted

fan for propulsion, vertical lift/landing and control. High aspect ratio (narrow) wings allow for 10 hours endurance like sail-plane flight efficiency.

Flying Elephant UAV

The Flying Elephant will take off from any place carrying up to a ton of cargo, fly silently and deposit the cargo wherever required. It will then independently take-off and fly to the next satisfied unit. The Jumbo UAVs can be used for military and civilian applications.

Miniature wireless sensors of the

above UAV are under evaluation in the laboratory.

Propulsion systems

The propulsion systems for UAVs depend on their payload, endurance and mission. Also, UAVs require reliable propulsion systems in order to achieve extended flight times with minimal quantities of fuel. Some of the propulsion systems are briefly described below.

Diesel and petrol engines. Both diesel and petrol engines are available of different cubic centimetres for compact systems. These are equipped with electronic controller, the sensors for the crankshaft position and inlet and cylinder head temperature, an ignition module, injector, fuel pressure accumulator and a fuel pump. Electronic fuel injection system, delivers an optimum fuel mix at all stages of flight, even for miniature aircraft.

Wankel engines. Because of the small size, weight, very low vibration and high power-to-weight ratio, Wankel engines are increasingly being used in UAV aircraft. The engine is approximately one third of the size and weight of a piston engine of equivalent

